

From Uncertainty to Determinism: Coarse-to-Fine Visual Floorplan Localization without Ray Matching

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Abstract

Visual Floorplan Localization (FLoc) has emerged as a promising solution for indoor localization by matching egocentric images against minimalist structural maps. However, due to cross-modal information asymmetry and repetitive indoor layouts, visual FLoc is fundamentally challenged by multimodal pose distributions, where visually identical observations map to distinct, spatially separated locations. Existing ray-matching-based methods tackle this by explicitly predicting sparse geometric or semantic rays, which inherently incurs information loss and demands resource-intensive preprocessing alongside exhaustive matching during inference. In this paper, we bypass the intermediate ray-matching paradigm and propose a coarse-to-fine visual FLoc framework that progresses from uncertainty to determinism. In the coarse stage, we design an image-conditioned pose diffusion model to parameterize the continuous multimodal pose distribution, effectively routing stochastically initialized pose particles toward distinct candidate modes. In the refinement stage, we propose a localized refiner that predicts bounded sub-meter pose residuals from candidate-centered floorplan crops, where structural ambiguities are largely eliminated. Our method effectively balances global multi-hypothesis tracking and local sub-meter refinement without requiring any offline map preprocessing or test-time lookup tables. Comprehensive results on the S3D (full) and ZInD benchmarks demonstrate that our approach achieves state-of-the-art accuracy and robustness.

Introduction

Visual Floorplan Localization (FLoc) (Chen et al. 2024) has emerged as a promising and scalable paradigm for indoor localization by aligning egocentric RGB observations with lightweight 2D floorplans, bypassing the need for storage-intensive 3D reconstructions (Liu, Li, and Dai 2017; Sarlin et al. 2019; Sattler, Leibe, and Kobbelt 2017) or large-scale image databases (Balntas, Li, and Prisacariu 2018; Arandjelović et al. 2016). Compared with dense 3D models, floorplans are lightweight, easy to maintain, and readily available in built environments, making visual FLoc particularly appealing for autonomous robot navigation (Li et al. 2025; Huang et al. 2025; Chen et al. 2026b) and augmented reality. However, the pronounced cross-modal information asymmetry between feature-rich visual observations and minimal-

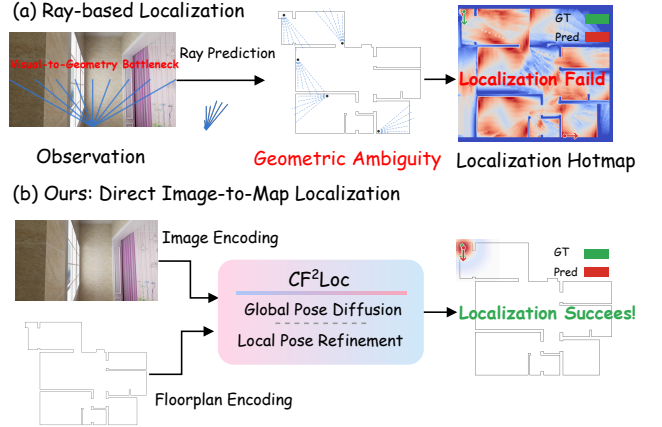


Figure 1: (a) Existing ray-matching-based methods rely on deterministic ray prediction and exhaustive map matching, which are susceptible to failure under cross-modal information asymmetry. (b) Revisiting visual FLoc from a pose distribution estimation perspective, our proposed CF²Loc framework progresses from probabilistic hypothesis generation to deterministic refinement, achieving robust global multi-hypothesis tracking and precise local refinement.

ist structural floorplans renders visual FLoc an inherently multimodal inference problem. Compounded by repetitive indoor layouts—such as symmetric rooms and long corridors—visually identical observations often map to distinct, spatially separated locations, yielding a complex multimodal pose distribution.

Early approaches (Min et al. 2022) addressed this challenge by performing handcrafted feature matching between visual inputs and local floorplan structures, requiring complex feature engineering and heuristics. Recent advances (Chen et al. 2024, 2025, 2026a; Ye et al. 2026) have shifted visual FLoc from front-end feature matching to back-end ray-based observation matching. Instead of directly estimating camera poses, these methods first compress observations into sparse geometric or semantic ray representations and localize them by matching against pre-computed map representations, as illustrated in Figure 1 (a). Despite their promising accuracy, ray-based methods adopt an indirect formulation for visual FLoc, which introduces three fundamental limi-

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tations. First, compressing feature-rich visual observations into sparse ray representations inevitably discards subtle appearance and structural cues, creating an information bottleneck prior to localization. Second, while ray matching can retain multiple candidate hypotheses at the map level, the front-end ray predictor deterministically maps observations to rays via a one-to-one function, failing to capture the multimodal pose distribution induced by cross-modal asymmetry. Finally, ray matching depends on map-specific ray databases that must be densely pre-computed, incurring heavy offline preprocessing overhead and requiring exhaustive test-time matching, thereby restricting scalability.

In this paper, we revisit visual FLoc from the perspective of direct pose distribution estimation. Our key insight is that visual FLoc should directly infer the conditional pose distribution over the map rather than reconstruct intermediate ray representations. Based on this insight, we propose a rendering-free, coarse-to-fine generative localization framework, named CF²Loc, which progresses from probabilistic hypothesis generation to deterministic refinement, as shown in Figure 1 (b). Specifically, in the coarse stage, an image-conditioned pose diffusion model parameterizes the continuous multimodal pose distribution across the entire floorplan, enabling effective global hypothesis generation without exhaustive map matching. Once coarse candidates are identified, localization is constrained to candidate-centered local floorplan crops, where structural ambiguities are largely eliminated. A lightweight local pose refiner then predicts bounded sub-meter pose residuals for precise local correction. By unifying generative global search with deterministic local refinement, our framework achieves robust global localization, accurate local refinement, and efficient inference without requiring offline map preprocessing or test-time lookup tables. Our contributions are summarized as follows:

(1) We formulate visual FLoc as a direct pose distribution estimation problem and propose CF²Loc, a rendering-free, coarse-to-fine generative localization framework that bypasses intermediate ray representations and exhaustive map matching.

(2) We introduce an image-conditioned pose diffusion model for continuous global hypothesis generation, alongside a localized pose refiner operating on candidate-centered floorplan crops for bounded sub-meter residual correction, establishing a seamless coarse-to-fine workflow.

(3) Extensive experiments on the Structured3D (full) and ZInD benchmarks demonstrate that our approach achieves State-Of-The-Art (SOTA) accuracy and efficiency while eliminating map-dependent preprocessing. The code will be publicly released.

Visual Localization

Visual localization estimates camera poses by establishing correspondences between query images and scene representations. Classical approaches rely on image retrieval (Arandjelović et al. 2016; Balntas, Li, and Prisacariu 2018), 2D–3D feature matching with Structure-from-Motion (SfM) models (Sarlin et al. 2019; Sattler, Leibe, and Kobbelt 2017), or direct pose regression using deep networks (Kendall, Grimes, and Cipolla 2015; Walch et al. 2017; Brachmann et al. 2017).

Despite notable advances, these methods depend heavily on appearance-rich scene representations—such as dense 3D reconstructions or extensive image databases—which are costly to construct, store, and maintain, and remain susceptible to environmental appearance changes. These limitations have motivated recent interest in visual FLoc, which localizes cameras by aligning egocentric observations with lightweight and readily available 2D floorplans.

End-to-End Visual FLoc

End-to-end visual FLoc directly learns a mapping from egocentric observations and floorplans to camera poses without explicit geometric matching. Representative methods, such as LaLaLoc (Howard-Jenkins, Ruiz-Sarmiento, and Prisacariu 2021), LaLaLoc++ (Howard-Jenkins and Prisacariu 2022), and PF-Net (Karkus, Hsu, and Lee 2018), learn joint image–floorplan representations for direct pose regression or particle-filter-based localization. While these approaches demonstrate that deep neural networks can capture cross-modal correspondences in an end-to-end fashion, they formulate localization as a deterministic mapping, implicitly assuming a one-to-one relationship between observations and poses. This deterministic formulation fails to account for the inherently multimodal nature of visual FLoc, where repetitive layouts and structural symmetries often produce multiple plausible pose hypotheses for a single observation. In contrast, our method directly models the conditional pose distribution via generative inference, effectively retaining multimodal pose hypotheses prior to deterministic local refinement.

Ray Matching-Based Visual FLoc

Recent visual FLoc methods (Chen et al. 2024; Wüest et al. 2026; Chen et al. 2025, 2026a; Grader and Averbuch-Elor 2025; Ye et al. 2026) formulate localization as a ray-based observation matching problem. Rather than directly estimating camera poses, these approaches first convert egocentric observations into sparse geometric or semantic ray representations and then determine camera poses by matching these rays against map-dependent rays rendered from floorplans. Subsequent works have extended this paradigm by incorporating depth uncertainty (Wüest et al. 2026), 3D geometric priors (Chen et al. 2025), semantic cues (Chen et al. 2026a), and probabilistic fusion strategies (Ye et al. 2026). Despite their promising results, ray-based methods solve localization indirectly through intermediate ray prediction and matching. This indirect formulation creates an information bottleneck prior to localization, overlooks the inherently multimodal pose distribution due to deterministic front-end ray prediction, and demands map-dependent ray pre-computation alongside exhaustive test-time matching. Furthermore, existing refinement strategies (Grader and Averbuch-Elor 2025) often rely on specific semantic constraints, limiting their general applicability across different floorplan representations. In contrast, our framework performs residual pose refinement directly in the pose space, making it applicable to both geometric and semantic floorplans while completely eliminating ray prediction, offline map preprocessing, and test-time lookup tables.

Preliminaries

Problem Formulation. Given an egocentric visual observation I and a 2D floorplan map M , the objective of visual FLoc is to estimate the camera pose $\mathbf{p} = (x, y, \theta)$, where (x, y) represents the 2D spatial coordinates and $\theta \in [0, 2\pi)$ denotes the orientation angle within the floorplan coordinate system. To accommodate the intrinsic ambiguities of indoor structures, we move beyond deterministic regression and instead formulate localization as estimating the continuous conditional pose distribution $p(\mathbf{p} | I, M)$. The optimal pose is theoretically defined by the maximum a posteriori (MAP) estimation:

$$\mathbf{p}^* = \arg \max_{\mathbf{p}} p(\mathbf{p} | I, M). \quad (1)$$

Directly optimizing Eq. (1) globally is computationally intractable due to the highly non-convex nature of the continuous pose space across a full map. We thus address this problem via a coarse-to-fine cascaded paradigm that decouples global uncertainty tracking from local deterministic refinement. In the first stage, we parameterize a global proxy distribution $q(\mathbf{p}) \approx p(\mathbf{p} | I, M)$ via a conditional generative model over the entire floorplan, from which a set of plausible candidate modes $\{\mathbf{p}_c^k\}_{k=1}^K$ is sampled. In the second stage, conditioned on a specific coarse candidate $\mathbf{p}_c$, we restrict the search space to a localized floorplan region $M_c \subset M$ centered at $\mathbf{p}_c$. The fine-grained localization is then formulated as estimating a bounded local pose residual $\Delta\mathbf{p} = (\Delta x, \Delta y, \Delta\theta)$ within a locally unimodal frame, yielding the final refined pose $\mathbf{p}_r = \mathbf{p}_c \oplus \Delta\mathbf{p}$, where $\oplus$ denotes the rigid body transformation composition operator.

Motivation. The pronounced cross-modal asymmetry between feature-rich visual observations and minimalist 2D floorplans renders visual localization a highly ill-posed problem. Widespread indoor structural symmetries and repetitive layouts inherently map a single observation to multiple disjoint locations, manifesting as a continuous multimodal posterior distribution $p(\mathbf{p} | I, M)$. Existing paradigms struggle with this multimodal nature. Deterministic regressors enforce a single-peak constraint on the output, leading to mode collapse and significant averaging errors in ambiguous regions. Conversely, SOTA ray-matching methods explicitly reconstruct 1D rays to preserve candidate hypotheses via exhaustive search; however, they suffer from an irreversible representation bottleneck during 2D-to-1D projection, alongside substantial test-time lookup table or rendering overheads during inference.

To circumvent these limitations, our core motivation is to bridge global multimodal uncertainty modeling with local unimodal determinism. We leverage a generative global search to parameterize the full posterior, capturing ambiguous hypotheses via non-parametric particle aggregation without exhaustive matching. Crucially, while structural ambiguities are prevalent globally, they are largely eliminated within a restricted local support. By adaptively cropping candidate-centered local maps, our framework transitions optimization from a complex multimodal search into a straightforward, unimodal residual regression, resolving global spatial ambiguities in a rendering-free manner.

Methodology

Global Pose Diffusion

Given I and M , the global diffusion model directly estimates the continuous conditional pose distribution across the entire floorplan. As illustrated in Fig. 2(a), the framework first extracts visual and floorplan features, performs cross-modal fusion to obtain observation-conditioned map representations, and models the multimodal pose distribution through conditional diffusion.

Multimodal Feature Encoding. The observation image is processed by a frozen visual backbone followed by lightweight projection and token-mixing layers, yielding dense visual tokens $\mathbf{F}_I \in \mathbb{R}^{N_I \times d}$ alongside a global image descriptor $\mathbf{g}_I$:

$$\mathbf{F}_I, \mathbf{g}_I = E_I(I). \quad (2)$$

The floorplan is rasterized into a structural tensor $\mathbf{X}_M$. For semantic floorplans, each semantic category is represented as an independent one-hot channel; otherwise, grayscale structural maps are used. A convolutional encoder extracts spatial map features, which are further augmented with learnable positional embeddings:

$$\mathbf{F}_M = E_M(\mathbf{X}_M) + E_{\text{pos}}(\mathbf{C}_M), \quad (3)$$

where $\mathbf{C}_M$ denotes the metric coordinates of map tokens.

To inject visual evidence into spatial map representations, we employ cross-attention with map tokens as queries and image tokens as keys and values:

$$\hat{\mathbf{F}}_M = \mathbf{F}_M + \text{CrossAttn}(\mathbf{F}_M, \mathbf{F}_I, \mathbf{F}_I), \quad (4)$$

yielding observation-conditioned floorplan tokens $\hat{\mathbf{F}}_M$, which preserve the full spatial support of the floorplan while encoding image-dependent localization cues.

Conditional Pose Diffusion. Rather than predicting intermediate ray representations, we directly model the conditional pose distribution in the continuous pose space. To circumvent boundary discontinuities associated with angular regression, the camera pose is parameterized as $\tilde{\mathbf{p}} = (x, y, \cos \theta, \sin \theta)$, where orientation is recovered using $\text{atan2}(\cdot)$ during inference.

Following the standard DDPM (Ho, Jain, and Abbeel 2020) formulation, Gaussian noise is progressively injected into the pose representation:

$$\mathbf{z}_t = \sqrt{\alpha_t} \mathbf{z}_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (5)$$

where $\mathbf{z}_0 = \tilde{\mathbf{p}}$.

A conditional denoising network predicts the injected noise from the noisy pose, diffusion timestep, global image descriptor, and observation-conditioned map tokens:

$$\hat{\epsilon} = \epsilon_\theta(\mathbf{z}_t, t, \hat{\mathbf{F}}_M, \mathbf{g}_I). \quad (6)$$

The diffusion model is optimized using the standard noise prediction objective:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t, \epsilon} [\|\epsilon - \hat{\epsilon}\|_2^2]. \quad (7)$$

Map-Pose Joint Rotation Augmentation. To prevent the diffusion model from overfitting to dataset-specific absolute orientation biases, we jointly rotate the floorplan and its corresponding ground-truth pose during training:

$$(M', \tilde{\mathbf{p}}') = R_i(M, \tilde{\mathbf{p}}), \quad i \in \{0, 1, 2, 3\}, \quad (8)$$

candidate pose errors:

$$s^* = \exp \left[-\frac{1}{2} \left(\frac{\|\mathbf{x}_c - \mathbf{x}^*\|_2^2}{\sigma_s^2} + \frac{\text{wrap}(\theta_c - \theta^*)^2}{\sigma_{s,\theta}^2} \right) \right], \quad (15)$$

where σ_s and $\sigma_{s,\theta}$ control translation and orientation error tolerances. The predicted confidence $\hat{s}$ is optimized using binary cross-entropy:

$$\mathcal{L}_{\text{score}} = -[s^* \log \hat{s} + (1 - s^*) \log(1 - \hat{s})]. \quad (16)$$

Coarse-to-Fine Inference

During inference, our framework executes visual FLoc in a coarse-to-fine cascaded manner by sequentially coupling global diffusion sampling with local deterministic refinement, as illustrated in Fig. 2(b).

Global Pose Hypothesis Generation. We first initialize N pose particles from isotropic Gaussian noise $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and iteratively apply the reverse diffusion process conditioned on the visual observation and floorplan features to yield a set of pose samples $\{\mathbf{p}^i\}_{i=1}^N$. These particles approximate the continuous multimodal pose posterior $p(\mathbf{p} \mid I, M)$, naturally capturing distinct plausible localization hypotheses. To extract representative pose modes from the sample set, we estimate particle density over the pose manifold via Kernel Density Estimation (KDE):

$$\mathcal{D}(\mathbf{p}) = \frac{1}{N} \sum_{i=1}^N \mathcal{K}_H(\mathbf{p} - \mathbf{p}^i), \quad (17)$$

where $\mathcal{K}_H(\cdot)$ denotes a product kernel in SE(2) space with bandwidth matrix H . The K density local maxima are extracted as coarse candidate modes $\{\mathbf{p}_c^k\}_{k=1}^K$.

Local Pose Refinement and Re-ranking. Each coarse candidate is processed independently by the local pose refiner. Specifically, $\mathbf{p}_c^k$, F_l , and M_c^k are fed into the refiner to predict local translation residuals $\Delta \hat{\mathbf{x}}^k$, orientation residuals $\Delta \hat{\theta}^k$, and hypothesis confidence scores $\hat{s}^k$. The refined pose is $\mathbf{p}_r^k = \mathbf{p}_c^k \oplus (\Delta \hat{\mathbf{x}}^k, \Delta \hat{\theta}^k)$. Finally, candidate hypotheses are re-ranked based on their predicted confidence scores:

$$k^* = \arg \max_{k \in \{1, \dots, K\}} \hat{s}^k, \quad (18)$$

and the top-ranked candidate $\mathbf{p}_r^{k^*}$ is output as the final localization pose.

Compared with existing ray matching-based paradigms, our inference pipeline solely relies on a few reverse diffusion steps and lightweight local refinement. It completely eliminates map-dependent ray rendering, test-time lookup tables, and exhaustive observation matching, thereby achieving both real-time execution and superior localization accuracy.

Experiments

Experimental Setup

Datasets. We evaluate our framework on two standard visual FLoc benchmarks: Structured3D (S3D) (Zheng et al. 2020) and the Zillow Indoor Dataset (ZInD) (Cruz et al. 2021), which provide complementary evaluation settings across synthetic and real-world scenes. S3D is a large-scale synthetic dataset containing 3,500 designer-created houses

with photorealistic rendered observations, ground-truth camera poses, and aligned 2D floorplans. Following prior protocols (Grader and Averbuch-Elor 2025; Ye et al. 2026), we conduct comparative experiments on the S3D (full) benchmark, comprising 3,296 fully furnished indoor scenes and 78,453 monocular perspective images. Monocular images are set to a horizontal field of view (FoV) of 80°, with a floorplan spatial resolution of 0.02 m under official train/validation/test splits.

ZInD is a real-world dataset comprising 1,575 unfurnished residential homes and 59,361 panoramic images paired with annotated floorplans. Following standard practice, panoramas are converted into perspective views for localization in the corresponding 2D floorplan coordinate frame.

Baselines. (1) **PF-net** (Karkus, Hsu, and Lee 2018) proposes a particle filter specialized for visual FLoc. Its observation model aims to learn the similarity between an image and the corresponding map patch.

(2) **MCL** (Dellaert et al. 1999) is the most popular framework for 2D localization on pure geometry maps.

(3) **LASER** (Min et al. 2022) represents the floorplan as a set of points and gathers the features of the visible points of each pose in the floorplan. It actively compares the rendered pose features with the query image features for visual FLoc.

(4) **F³Loc** (Chen et al. 2024) is a classic visual FLoc method that proposes a probabilistic model consisting of a ray-based observation module and a histogram filtering module.

(5) **3DP** (Chen et al. 2025) injects 3D geometric priors into visual FLoc which significantly improve the single-frame and multi-frame FLoc accuracy without the need of any semantic labels.

(6) **RSKFLoc** (Chen et al. 2026a) serves as a strong baseline, incorporating unsupervised learning-based room style knowledge into visual FLoc without requiring explicit semantic information.

(7) **SemRayLoc** (Grader and Averbuch-Elor 2025) represents a strong baseline in semantic FLoc. It utilizes sparse semantic floorplan priors to predict semantic rays, generating structural-semantic probability volumes to improve localization performance.

(8) **SceneAligner** (Cho, Cai, and Averbuch-Elor 2026) projects 3D reconstructions onto dense maps and aligns them with floorplans to achieve the previous SOTA in semantic-free visual FLoc.

(9) **FoD** (Ye et al. 2026) proposes a probabilistic fusion strategy that achieves state-of-the-art visual FLoc performance by fusing depth and semantic information. FoD_d is a variant of FoD, indicating that the floorplan semantics is used during training, while only the depth is predicted during inference.

Evaluation Metrics. In accordance with standard protocols (Chen et al. 2024; Grader and Averbuch-Elor 2025; Ye et al. 2026), we report recall metrics at spatial accuracy thresholds of 0.1 m, 0.5 m, and 1.0 m. Additionally, we report joint pose recall requiring spatial accuracy within 1.0 m and orientation error bounded within 30°. Recall measures

the percentage of query images localized within the specified error thresholds.

Implementation Details. Our framework is implemented in PyTorch and trained on a single NVIDIA RTX 5880 Ada GPU. The visual observation encoder utilizes the frozen ViT-S backbone from Depth Anything V2 (Yang et al. 2024). Geometric floorplans are formatted as single-channel grayscale rasters, whereas semantic floorplans are represented as five-channel one-hot tensors. The floorplan encoder comprises a convolutional stem and the first two stages of ResNet-18, followed by two residual context blocks. The global diffusion denoising network incorporates two pose-to-map attention blocks (Vaswani et al. 2017). All attention mechanisms employ four heads with a dropout rate of 0.1.

The diffusion model is optimized via AdamW (Loshchilov and Hutter 2017) using a cosine noise schedule ($T = 1000$), a batch size of 8, an initial learning rate of 10^{-4} , a minimum learning rate of 10^{-5} , and a weight decay of 10^{-4} . Cosine annealing and gradient clipping (maximum norm 1.0) are applied throughout training. Non-semantic and semantic models are trained for 30 and 60 epochs, respectively. Joint map-pose rotation augmentation is applied with probability 0.5 using discrete rotations from $\{90^\circ, 180^\circ, 270^\circ\}$.

The local pose refiner reuses the frozen visual encoder and token-mixing layers from the diffusion stage. For each coarse candidate, a $5\text{ m} \times 5\text{ m}$ oriented local floorplan crop is extracted and resized to 256×256 . Refiner networks are optimized via AdamW for 30 epochs with a batch size of 16. Training candidates are generated via multi-scale pose perturbation: 50% sampled from fine Gaussian noise ($0.3\text{ m}/10^\circ$), 30% from moderate noise ($0.8\text{ m}/25^\circ$), and 20% from uniform noise within $1.5\text{ m}/45^\circ$. Loss hyperparameters are set to $\sigma_h = 0.2\text{ m}$, $\sigma_s = 0.5\text{ m}$, $\sigma_{s,\theta} = 20^\circ$, $\lambda_\theta = 1.0$, $\lambda_h = 1.0$, and $\lambda_s = 0.25$.

During inference, 64 pose particles are initialized from Gaussian noise and refined through 10 reverse diffusion steps. Particle density is estimated using an SE(2) kernel with spatial and angular bandwidths of 0.75 m and 20° , respectively. K candidate modes are independently refined, and the hypothesis with the highest predicted confidence is selected as the final camera pose.

Comparative Studies

Comparison with SOTA Methods. Table 1 and Table 2 compare our method with SOTA visual FLoc approaches on the S3D (full) and ZInD benchmarks. When training without semantic floorplans, our complete framework consistently achieves top-performing results across all metrics. On S3D (full), it substantially improves upon the previous best 3D reconstruction-based baseline, SceneAligner, elevating the recall from 3.5%, 37.5%, 53.8%, and 51.6% to 11.3%, 64.4%, 73.7%, and 72.6% under the 0.1 m, 0.5 m, 1.0 m, and 1.0 m/30° criteria, respectively. On the real-world ZInD benchmark, our method similarly outperforms existing approaches by a wide margin, boosting the 0.5 m recall from 11.1% (LASER) and 7.9% (F³Loc) to 45.5%, while reaching 57.3% and 45.9% under the 1.0 m and 1.0 m/30° metrics, respectively. These marked gains verify that directly parame-

Method(Venue)	Sem.	0.1 m	0.5 m	1 m	1m 30°
<i>Training without semantic floorplans</i>					
PF-net(CoRL 2018)	No	0.2	1.3	3.2	0.9
MCL(ICRA 1999)	No	1.3	5.2	7.8	6.4
LASER(CVPR 2022)	No	0.7	6.4	10.4	8.7
F ³ Loc(CVPR 2024)	No	1.5	14.6	22.4	21.3
RSKFLOC(AAAI 2026)	No	4.5	31.7	38.7	37.5
3DP(ACM MM 2025)	No	4.8	32.8	40.3	39.0
SceneAligner(Arxiv 2026)	No	3.5	37.5	53.8	51.6
CF ² Loc w/o Sem.	No	6.4	58.9	73.4	72.3
CF ² Loc w/o Sem. + Refine	No	11.3	64.4	73.7	72.6
<i>Training with semantic floorplans</i>					
SemRayLoc(ICCV 2025)	Yes	5.7	45.5	58.8	57.5
FoD _d (CVPR 2026)	No	11.4	56.4	61.4	60.8
FoD(CVPR 2026)	Yes	12.0	65.2	71.9	71.4
CF ² Loc w/Sem.	Yes	8.0	65.2	78.7	78.1
CF ² Loc w/Sem. + Refine	Yes	15.0	71.0	79.1	78.4

Table 1: Comparative experimental results on the S3D (full) dataset. Sem. indicates whether to predict semantic rays during inference. FoD_d is a variant of FoD, indicating that the floorplan semantics is used during training, while only the depth is predicted during inference.

Method(Venue)	Sem.	0.1m	0.5m	1m	1m 30°
<i>Training without semantic floorplans</i>					
LASER(CVPR 2022)	No	1.4	11.1	17.6	13.6
F ³ Loc(CVPR 2024)	No	0.7	7.9	15.1	11.5
CF ² Loc w/o Sem.	No	3.3	36.3	55.5	44.6
CF ² Loc w/o Sem. + Refine	No	9.7	45.5	57.3	45.9
<i>Training with semantic floorplans</i>					
SemRayLoc(ICCV 2025)	Yes	3.3	26.6	38.0	31.9
FoD _d (CVPR 2026)	No	5.3	31.1	39.3	35.4
FoD(CVPR 2026)	Yes	8.1	43.6	53.5	50.2
CF ² Loc w/Sem.	Yes	4.3	43.4	61.9	53.9
CF ² Loc w/Sem. + Refine	Yes	11.5	51.7	63.2	54.7

Table 2: Comparative experimental results on the ZInD dataset. Sem. indicates whether to predict semantic rays during inference.

terizing the conditional pose distribution and refining candidate modes is significantly more effective than deterministic ray prediction and exhaustive observation matching, particularly in the absence of semantic cues.

When training with semantic floorplans, our method continues to establish new SOTA performance. Compared with the strongest semantic baseline, FoD, our framework improves recall on S3D (full) from 12.0%, 65.2%, 71.9%, and 71.4% to 15.0%, 71.0%, 79.1%, and 78.4% across the four evaluation criteria, respectively. On ZInD, it further elevates the 0.5 m, 1.0 m, and 1.0 m/30° recalls from 43.6%, 53.5%, and 50.2% to 51.7%, 63.2%, and 54.7%, respectively. These results underscore the clear advantage of direct multi-modal pose distribution modeling over indirect ray-matching paradigms. Furthermore, the proposed local refiner consistently yields notable accuracy boosts under the stringent

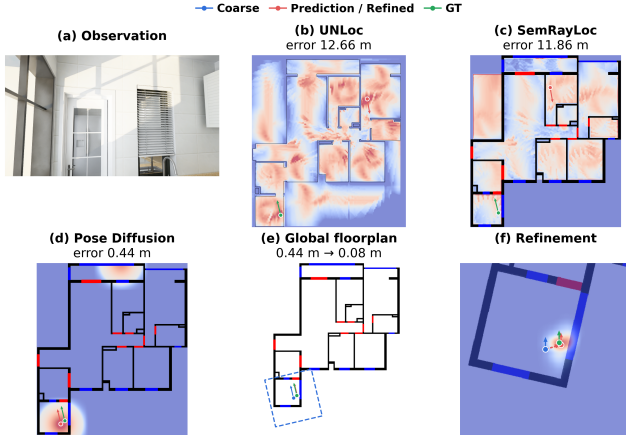


Figure 3: A qualitative comparison of our method (d-f) with existing methods (b-c) on S3D (full). F³Loc and Unloc do not use semantics, while SemRayLoc does.

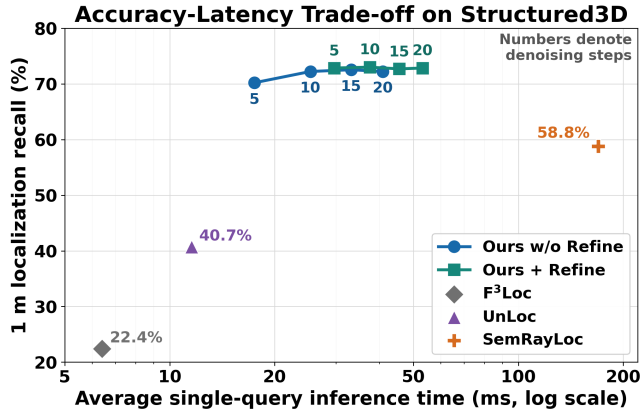


Figure 4: Trade-off analysis comparing visual FLoc performance (1.0 m Recall) and computational latency (ms) across varying denoising steps. Our method achieves superior accuracy while remaining computationally efficient.

0.1 m criterion (e.g., 12.0% $\rightarrow$ 15.0% on S3D (full) and 8.1% $\rightarrow$ 11.5% on ZInD), while fully preserving the robust global localization capacity established by the diffusion stage. The consistent superiority across both synthetic and real-world benchmarks demonstrates the effectiveness and strong generalization ability of our coarse-to-fine generative framework.

Figure 3 presents a qualitative comparison between our method and representative baselines. While methods such as F³Loc, UnLoc, and SemRayLoc frequently suffer from catastrophic mode mismatches in symmetric indoor layouts, our approach reliably identifies true candidate modes through generative global search. The local refinement stage subsequently sharpens pose accuracy, yielding significant gains under strict error thresholds.

Accuracy-Efficiency Trade-off. Figure 4 illustrates the trade-off between localization recall and inference latency under varying diffusion denoising steps. Our method

MS	RA	Refiner	LPR	0.1 m	0.5 m	1 m	1 m 30°
-	-	-	-	4.2	45.8	63.7	62.7
✓	-	-	-	5.4	53.8	71.5	70.7
✓	✓	-	-	6.3	58.5	73.2	72.2
✓	✓	✓	-	6.9	59.6	73.1	72.0
✓	✓	✓	✓	11.3	64.4	73.7	72.6

Table 3: Ablation studies of our proposed method on the S3D (full) dataset. MS, RA, and LPR denote Mode Selection, Rotation Augmentation, and Local Patch Rotation, respectively.

Local Map Size	0.1 m	0.5 m	1 m	1 m 30°
3 m $\times$ 3 m	9.7	62.2	73.3	72.3
5 m $\times$ 5 m	11.3	64.4	73.7	72.6
7 m $\times$ 7 m	10.3	63.1	72.0	71.0

Table 4: Results of parametric studies on the local map size.

achieves over 70% recall at 1.0 m using only 5 denoising steps with low runtime overhead, and converges rapidly around 10 steps, beyond which further iterations yield diminishing returns. This confirms that accurate conditional pose distribution modeling can be achieved with minimal generative sampling steps. Furthermore, compared with ray matching-based methods, our framework achieves a markedly superior accuracy-latency trade-off. Specifically, it improves the 1.0 m recall from 58.8% (SemRayLoc) to over 73% while reducing inference time by nearly 4 $\times$. These results demonstrate that direct pose distribution estimation enables both SOTA localization precision and real-time inference without requiring offline map preprocessing or online ray matching.

Ablation Studies

Table 3 presents comprehensive ablation studies conducted on S3D (full). Starting from the baseline diffusion model, incorporating candidate mode selection substantially improves the 0.5 m recall from 45.8% to 53.8% and the 1.0 m recall from 63.7% to 71.5%. This confirms the necessity of preserving dominant mode hypotheses in a multimodal pose distribution rather than naively averaging diffusion samples. Applying joint map-pose rotation augmentation further boosts the 0.5 m recall to 58.5% and the 1.0 m recall to 73.2%, indicating enhanced robustness to arbitrary global floorplan orientations. Integrating the local refiner consistently improves fine-grained localization accuracy, with gains becoming significantly more pronounced after enabling local patch canonicalization (rotation alignment). Specifically, 0.1 m recall jumps from 6.9% to 11.3%, and 0.5 m recall improves from 59.6% to 64.4%, while 1.0 m recall remains largely stable. This highlights that local patch canonicalization effectively normalizes candidate-local coordinate frames, allowing the refiner to concentrate on regressing bounded residual displacements. Overall, these progressive gains validate that each component provides complementary advantages, collectively enabling robust global mode tracking and precise local refinement.

Parametric Studies

We conduct sensitivity analyses on the local map size. As shown in Table 4, a $5\text{ m} \times 5\text{ m}$ crop yields the best accuracy by balancing spatial layout context with local unimodality—smaller patches lack context, whereas larger ones reintroduce structural ambiguities. Furthermore, $K = 1$ (selecting the top-confidence candidate) suffices to achieve top-tier performance on both S3D (full) and ZInD, confirming the reliability of our diffusion sampling and confidence scoring in identifying true pose modes.

Conclusion

In this paper, we revisit visual floorplan localization (FLoc) from the perspective of direct conditional pose distribution estimation and propose a rendering-free, coarse-to-fine generative framework. Rather than relying on intermediate 1D ray prediction or exhaustive observation matching, our method parameterizes the full multimodal pose posterior via an image-conditioned pose diffusion model, followed by a lightweight local refiner for precise residual regression. This coarse-to-fine formulation seamlessly bridges global uncertainty modeling with local unimodal determinism, enabling robust multi-hypothesis tracking and fine-grained pose estimation within a unified architecture applicable to both geometric and semantic floorplans. Extensive evaluations on the S3D (full) and ZInD benchmarks demonstrate that our approach consistently establishes new SOTA accuracy while supporting real-time inference without requiring offline map preprocessing. We hope this work provides a scalable, rendering-free alternative to traditional ray-matching paradigms and inspires further research in generative cross-modal spatial reasoning.

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